AI Learning Companion

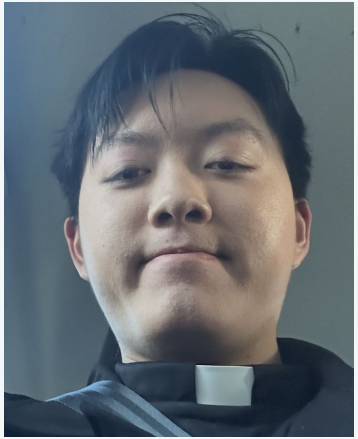
Team 7 POVs, HMWs, Brainstorming & Experience Prototyping

Team Members

Ayala Wang

‘26

San Jose, CA



Anthony Wong

‘26

Sacramento, CA

Junjie Liu

‘26

Las Vegas, NV

Quick Recap

Previously…

- We covered some key learnings from our initial interviews
- Brief summary of next week's plan

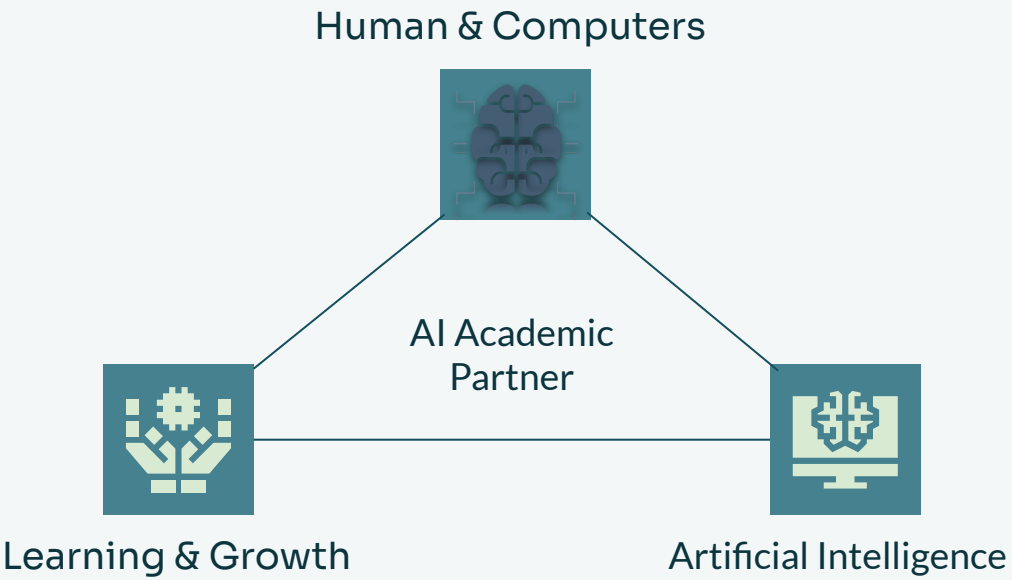
This time…

- We will refine our problem domain
- Analyze new interview information
- Generate POVs, HMWs. Propose solutions & experience prototypes

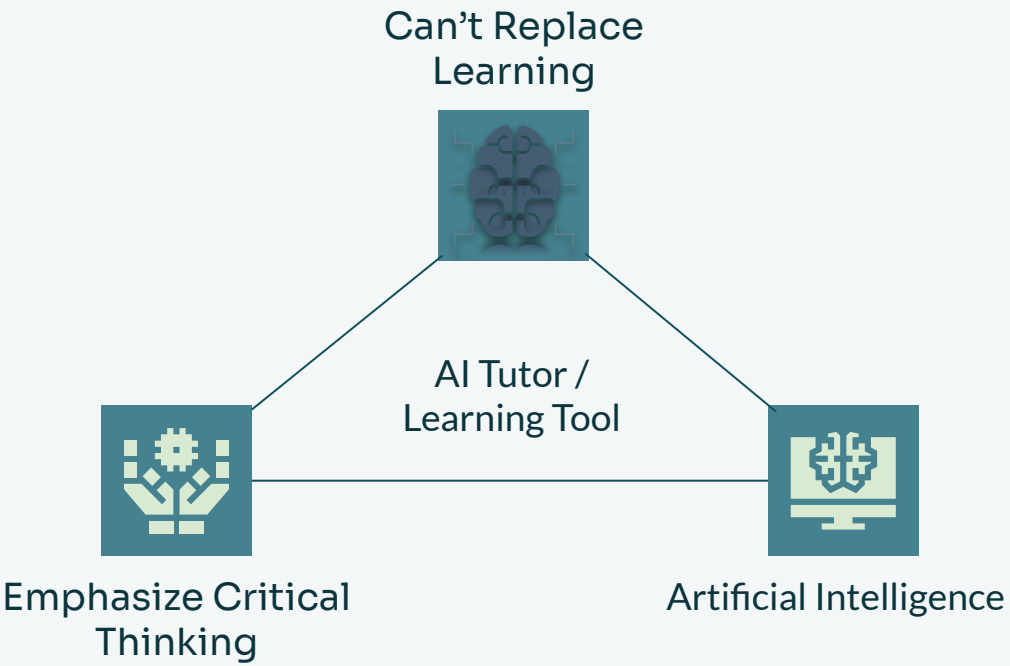
01

Problem Domain

Old Problem Domain



New Problem Domain



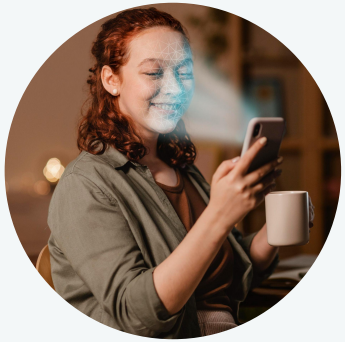
02

Additional Needfinding

Old Interviewees

Person A

53 year old Jiu-Jitsu Professor



Person B

21 year old college student at SBCC

Person C

Junior Software Engineer at a learning company

New Interviewees

Person D

Math Teacher with 24 year old sibling in grad school

Person E

21 year old college student at SBCC going to med school

Previous Insights

General Consensus from interviewees:

- AI is helpful to kickstart learning in the early stages
- But should not be relied on as a primary source of information but only as an assistant instead
- Interviewees emphasized that AI should not replace learning

Person D

SAYS

Observations: What did this person SAY that surprised you?

- "I give them hints but not detailed solutions"
- "For abstract mathematics questions, I discuss a concrete example"
- "AI is beneficial as long as it's used wisely"
- "They may use AI for references, but they must engage in critical thinking"
- "They should be able to complete closed-book exams independently"
- "Teachers should be aware of AI tools and help students use them appropriately"
- "Some schools are reluctant to adopt AI"
- "There are many more open-access publications now"
- "I support AI use, but it shouldn't replace thinking"
- "There are many more open-access journals and resources now than before""

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks real understanding comes from working through difficulty
- Giving answers too quickly hurts long-term learning
- Thinks AI can support learning, but it can't replace reasoning
- Thinks students must maintain intellectual independence
- Over-reliance on AI weakens mastery
- Educators should shape how AI is used not ignore it
- Struggle is a necessary part of graduate education
- Efficiency is less important than depth of understanding
- Students should be able to perform without external tools
- Academic rigor must be protected as technology evolves

DOES

Observations: What did this person DO that surprised you?

- Intentionally avoid giving full solutions
- Guides by pointing key observations
- Relies on examples to help build intuition
- Shares own graduate school experiences when advising
- Allows struggle before stepping in
- Allows AI usage only within clear boundaries
- Emphasizes independent problem solving
- Encourages to verify information themselves
- Trusts traditional exams to test understanding
- Mentors informally rather than micromanaging

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Cautiously optimistic about AI's role in education
- Protective of academic standards
- Responsible for modeling good learning practices
- Concerned about students becoming dependent on tools
- Confident in traditional teaching methods
- Measured and deliberate when discussing AI
- Slightly frustrated by institutional hesitation around AI
- Calm about technological change, but vigilant
- Motivated to preserve deep learning
- Trusting of students, but aware of potential shortcuts

Person E

SAYS

Observations: What did this person SAY that surprised you?

- "She's pretty independent at this stage and manages most of her academics on her own"
- "Her studying is very self-directed"
- "It's less homework and more constant studying and revision"
- "The material is extremely content-heavy"
- "She usually tries to work through things herself first"
- "Sometimes she just vents when the workload feels overwhelming."
- "It's demanding and stressful, but also very structured."
- "Time management and burnout are the biggest challenges."
- "At first, I was unsure about AI because accuracy really matters in medicine."
- "I see AI as more of a supplement - like a quick tutor"

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks deep learning still requires effort and repetition
- Thinks AI can help with understanding but not mastery
- Accuracy matters more than speed in medicine
- Burnout is a constant risk
- Studying well is about structure and discipline
- Short explanations can save time and reduce frustration
- Real learning can't be outsourced
- Efficiency is helpful but only if learning quality stays high
- Support systems matter even for independent learners
- AI should feel like tutoring not automation

DOES

Observations: What did this person DO that surprised you?

- "She studies for long hours reviewing lectures and materials"
- "She relies heavily on practice questions and question banks."
- She uses textbooks and lecture slides before outside tools
- She reaches out to classmates or professors when stuck.
- She explains concepts out loud to process them.
- She uses AI to clarify or simplify difficult topics.
- She avoids using AI as a shortcut.
- She manages most studying independently.
- She vents stress verbally rather than asking for answers
- She balances multiple resources instead of relying on one

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Overwhelmed by workload intensity
- Stressed but highly motivated
- Reassured by structured routines
- Frustrated by time pressure
- Cautiously optimistic about AI
- Relieved when concepts are clarified quickly
- Mentally fatigued from constant studying
- Confident in her independence
- Supported by having someone to talk to
- Anxious about burnout and long-term sustainability

03

POV Development & HMWs

Person A
Location: Online
UserType: Average User:
Interviewers: Ayala (Interviewer) & Anthony (Scribe/Recording)

Point of View

We met ...

- An assistant professor in the Department of Mathematics in the University of Manitoba

We were surprised to to realize that...

- Even at the graduate level, learning still relies heavily on informal guidance—sharing hints, concrete examples, and personal experience—rather than step-by-step solutions, and that AI is viewed as helpful but only when it reinforces independent thinking.

We wonder if this means...

- AI's real value in advanced education isn't replacing problem-solving, but acting as a reflective partner that supports intuition-building and exploration while preserving rigor and assessment integrity.

It would be game-changing...

If AI tools were intentionally integrated by educators as guided reference and thinking aids—helping students test ideas, access research, and develop insight—while still ensuring they can perform independently in closed-book, high-stakes settings.

How might we ...

instill confidence and self-trust in graduate students' problem-solving?

reduce the mental load of tackling abstract or open-ended math problems?

reintroduce curiosity and excitement into parts of learning that feel intimidating?

support moments that trigger anxiety, like proofs, exams, or research uncertainty?

help students safely expand their intellectual comfort zone?

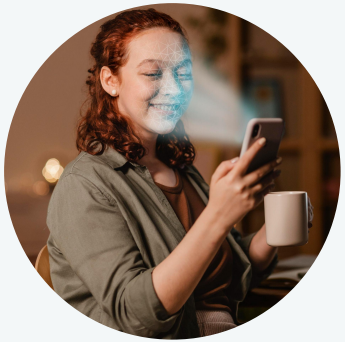
make learning environments feel grounding, reflective, and focused rather than high-pressure?

make boundaries around academic integrity and responsible AI use more intuitive?

make students feel in control of their learning even when concepts are unfamiliar?

make AI feel like a trusted academic space, not a shortcut or threat?

help students reframe "high-stakes" tasks (e.g., proofs, presentations, exams) as learnable processes?



Person B
Location: Online
UserType: Average User:
Interviewers: Anthony (Interviewer) & Junjie (Scribe/Recording)

Point of View

We met ...

- Person B, a 21 year old college student who frequently encounters AI in education and feels cautious towards it

We were surprised to to realize that...

- At a college level, AI is already widespread. While using it she has to actively do her best to utilize her own critical thinking to ensure she actually retains information while using these AI tools. Feels compelled to use it to keep up.

We wonder if this means...

- Students are currently being forced to survey their own learning process, ensuring the tools don't take over their learning in exchange for speed and convenience

It would be game-changing...

- we help students use AI in a way that supports their learning while preserving critical thinking and proper understanding

How might we ...

How might we help students use AI without replacing critical thinking

How might we prevent AI from just giving students the answers

What methods are there to force students to have active participation

What methods are there to support long term retention?

What can we do to help students keep up with other students who use AI w/o restraint

How can we keep the speed of AI while maintaining the same level of learning

How do we cap AI to avoid it from just giving the result / answers

How do we support students without them from becoming over reliant on the AI tool

How might we design a tool that may withhold hints/answers from its user

How can we reduce the anxiety that comes from imposter syndrome and the feeling of being left behind

Person C
Location: Zoom
UserType: Average User:
Interviewers: Junjie (Interviewer) & Anthony (Scribe/Recording)

Point of View

We met ...

- Person C, a junior software engineer at a learning-focused company with a few years of experience

We were surprised to to realize that...

- Despite the rapid adoption of AI, professional learning still relies on largely official documentations, hands-on experimentation, and learn directly from experienced peers. AI often becomes unreliable as problems move beyond surface level understanding

We wonder if this means...

- AI's true value in learning may not be in providing direct answers or replacing critical thinking, but in acting as an entry point that helps learners orient themselves, ask better questions, and build foundational understanding.

It would be game-changing...

- AI tools were intentionally designed as lightweight, guided learning partners that offer hints, prompts, and references to support sense-making—while preserving the learner's ability to reason independently and perform without AI support in high-stakes or closed-book contexts.

How might we ...

design AI that offers small hints or “thinking crumbs” instead of complete answers?

help learners use AI as a starting point without becoming dependent on it?

support deep, domain-specific learning where AI is more prone to error?

encourage users to validate and question AI-generated information rather than trusting it automatically?

help learners recognize when to stop using AI and continue reasoning independently?

scaffold learning in a way that mirrors how professionals learn through documentation, experimentation, and peer discussion?

design AI interactions that promote active problem-solving rather than passive consumption?

balance productivity gains from AI with long-term skill development and understanding?

build trust in AI learning tools through transparency, sources, and explainability rather than authority?

integrate AI into learning workflows without disrupting existing, effective learning practices?

How might we help professionals learn faster with AI while still requiring them to do the thinking themselves?

Top 3 HMWs

Person A
make boundaries around academic integrity and responsible AI use more intuitive?

Person B
How might we help students use AI without replacing critical thinking

Person C
How might we help professionals learn faster with AI while still requiring them to do the thinking themselves?

04

Solutions

HMW 1

Color label tasks (green/yellow/red)

Ask students why they are using AI before giving help

Unlock answers only after students attempt the task first

Prevent AI help on tasks the mimic exams

Built in reflection prompts. Go back to check retention/summarize

Let students witch between study, practice, or submission modes

Require Students to explain concepts before AI provides answers

Should we just disable AI as a whole to force people to go back to age w/o AI. We could totally pull a 30k and erase all existence of AI

Offer prompts designed to promote real understanding

Show AI's confidence or uncertainty for each response

Only provide citations, not full explanations

Ask students to relect on what they learned after using AI

Let professors set rules on how AI can be used

Give a summary of how AI was used for transparency

Adjust AI support based on workload or signs of burnout

HMW 2

Gate kept solutions, must explain / send in a pdf before extra info given

Utilize a bunch of active call back techniques. Learning, questions, summaries, etc.

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Require an attempt by the student first. ONLY GIVE PARTIAL HINTS otherwise

Built in reflection prompts. Go back to check retention/summarize

Have an AI that double checks your reasoning

INCLUDE REFLECTIONS, require an exit ticket type of answer

Should we just disable AI as a whole to force people to go back to age w/o AI. We could totally pull a 30k and erase all existence of AI

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Explain it back features?

For mental aspect maybe include progress? Encourage learning

Reassurance measures. Let them know they aren't behind

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Maybe let the person control strictness?

Modes of restrictions to allow learning?

HMW 3

Forces users to attempt before progressing, only give them piece by piece information.

Before the AI gives a hint, the user must explain their current understanding in their own words; the AI then adapts hints based on gaps in reasoning.

Instead of answering questions, the AI identifies and challenges hidden assumptions in the user's approach ("What are you assuming about X here?").

An AI that never explains directly, but instead guides users to the most relevant sections of official docs and asks questions that help interpret them.

Built in reflection prompts. Go back to check retention/summarize

AI only activates after a task is completed, helping users reflect on decisions, trade offs, and alternative approaches they could explore next time.

Hints are time-locked or attempt-locked; the AI nudges users to keep working before offering help, reinforcing persistence.

The AI remains silent until the user explicitly logs what they tried, what failed, and why — only then does it offer directional feedback.

The AI helps by adding constraints rather than removing them. For example: "Solve this without using X," or "What if X increased by tenfold?

Instead of giving correct solutions, the AI generates plausible wrong approaches and asks the user to critique them.

Limited hints given per problem, e.g. 3 tries before the user is on their own

Instead of giving answers, search the most relevant tutorials/learning resources available.

Once assistance is given for a certain type of problem, no assistance is allowed on another of the same type.

The AI asks users to rate their confidence before receiving help and adjusts its responses accordingly. lower confidence = broader guidance, higher confidence = sharper challenges.

Lightweight AI that only gives limited hints. Further requests are denied and links to other resources are given

Top 3 Solutions

Solution A

AI adapts its support based on a student's workload and signs of stress, giving extra guidance when needed without replacing independent learning.

Solution B

Depending on what the user needs, there can be varying levels of AI involvement. Either step by step guide while learning vs sparse hints while practicing

Solution C

Before the AI gives a hint, the user must explain their current understanding in their own words; the AI then adapts hints based on gaps in reasoning.

05

Experience Prototypes

Solution A Prototype

Critical Assumption:

- AI that adapts its support based on workload or signs of burnout will help students stay engaged, reduce stress, and maintain learning without replacing independent problem-solving.

Roles:

- Participant: College Students
- How Recruited: Recruited via personal network - classmates' siblings
- Why Relevant: These students have heavy workloads, use AI occasionally for study support, and are familiar with the challenge of AI sometimes doing too much for them.

Description:

- The participant is given a study task or set of practice problems.
- The participant can request AI help at any time.
- The facilitator monitors workload indicators (time spent on task, number of attempts, self-reported stress).
- When workload or stress is detected as high:
  - The facilitator provides extra guidance, such as simplified explanations, directional hints, or review prompts.
  - No full answers are given.
- When workload/stress is low:
  - The facilitator provides standard clarification or references only.
- The participant continues working using the adaptive support.
- At the end, the participant reflects on the experience.

Solution A Results

What Worked:

- Participants felt less stressed when AI provided extra guidance during high-workload moments.
- Hints and simplified explanations helped them continue working without feeling stuck.
- Students reported maintaining control over learning, not feeling replaced by AI.
- Reflection prompts helped reinforce learning and awareness of progress.

What didn't work:

- Monitoring workload and stress required manual facilitation, which might be harder to scale digitally.
- A few participants wanted more concrete feedback, showing a tension between guidance and independence.

New Learnings

- Adaptive AI can reduce cognitive overload while preserving student agency.
- Students appreciate flexible control over AI strictness and timing.
- Reflection and prompts are essential to make adaptive support effective.
- Implementation must balance support vs. giving too much, otherwise learning engagement can drop.

Validity and going forward

- The prototype validates the assumption that workload-aware AI helps without replacing independent thinking.
- Next steps: implement automated monitoring of workload/stress, test with a larger user group, refine the level of hints provided.

Impact: 7/10: This solution directly addresses key user pain points: heavy workloads, stress, and risk of AI replacing independent learning. Not impactful for some though

Reach: 8/10: Almost any student or learner who uses AI could benefit from this system. However some do not need it as much as others

Novelty: 9/10: While adaptive AI exists in some learning platforms, tailoring AI specifically to workload and burnout indicators is less common.

Solution B Prototype

Critical Assumption:

- Depending on where someone is in the learning process, they need different amounts of aid.
-
-

Participants:

- Participant: College Students
- How Recruited: Found among classmates in CS class
- Why Relevant: Many of these students have already encountered/familiar with using AI for education and have likely ran into the issue of having it complete too much

Description:

- The participant is given a short leetcode problem
  - Attempted to choose one out of the scope of the student
- Instead of looking up solutions/additional aid, they were allowed to ask us proctors or ChatGPT for hints about the questions
- If it was too difficult for them they could ask ChatGPT to explain the answer
- After 30 minutes working on the problem we could gauge their understanding of the problem in comparison to the start

Solution B Results

What Worked:

- Participants who asked for hints made considerable progress. They also had a good understanding of the problem and the topics involved
- Avoided the issue of simply copying of the solution

What didn't work

- Most of the participants just asked for the step by step guide from GPT after 5-10 minutes.
- GPT was too readily available
- Hints couldn't aid the people who didnt understand the fundemental

New Learnings

- Difficulty needs to be properly measured
- Hints are usually more effective at active recall if the participants are willing
- If outside of a person's scope, straight tutorials/step by step instructions may be necessary

Validity and going forward:

- The solution technically works in teaching people
- Still feels very self guided in terms of the student managing their own learning to ensure active recall

Impact:6/10, if the participant doesn't manage their own learning then it can have very little long term retention

Reach: 10/10, can work for any level due to the broadness of the aid given

Novelty: 8/10, rarely see levels of AI aid outside of a couple tools

Solution C Prototype

Critical Assumption:

- Requiring users to explain their current understanding before receiving AI help improves learning and maintains user agency without feeling frustrating or overly burdensome.

Participants:

- College Students
- How Recruited: Found among classmates in CS / Math classes
- Why Relevant: They are familiar with AI, and have used it for education/learning, are more likely to provide good feedback.

Script:

- The participant is given a short task
- The participant is told they may ask the “AI” for help at any time.
- When the participant asks for help, the facilitator responds:
- “Before I can help, please explain what you currently understand about the problem.”
- The participant writes or verbally explains their understanding.
- The facilitator reviews the explanation and responds only with:
    - Clarifying questions
    - Identification of gaps or misconceptions
    - Directional hints or prompts (no answers or solutions)
- The participant continues working on the task for several minutes using the guidance.
- At the end, the participant reflects on the experience.

Solution C Results

What Worked:

- Requiring users to explain their thinking increased engagement and focus
- Participants were able to identify gaps in their own understanding while explaining
- The interaction felt closer to peer discussion than AI assistance

What didn't work

- The explain-back step initially felt slower than immediately receiving help
- Without clear guidance, explanations sometimes became too vague

New Learnings

- Articulation itself is a powerful learning moment, even before hints are given
- Users are more willing to accept friction when it clearly improves understanding

Validity and going forward

- Impact: 8/10
- Reach: 10/10
- Novelty: 7/10
- Results suggest the explain-back mechanism effectively preserves agency while supporting learning.

Chosen Solution

Solution C: Explain-to-Unlock AI

Why: Had the most impact, forcing the users to actively recall information, resulting in greater retention

Who:

- Its use as an AI learning tool makes it broad in its populations
- Works best for **young adults and high school students** where most of the work requires critical thinking

Who's Left Out:

- Young children may get less usage due to the lack of direct explanations. May be difficult for them to articulate their thoughts and misunderstandings

Next Steps:

- Serves a general learner audience
- May leave out younger children
- Refine on solution if needed
- Build website for proposed solution

06

Appendix

Person A

SAYS

Observations: What did this person SAY that surprised you?

- "Just as an assistant. To help with schedule, write a professional email, presentation."
- "When I first saw AI, I felt like people were cheating."
- "If it's used as an assistant, I think it's valid."
- "Parents are the ones they go to when they need help."
- "AI should not be the main source of information."
- "I don't see AI being used in my child's education yet."
- "AI is useful, but only if you still think for yourself."
- "It should only help, not do the work"
- "People use AI to finish faster instead of learning."
- "We use AI for things like schedules or presentations."

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Struggling first is an important part of real learning
- Confidence matters more than finishing an assignment quickly
- Help should scaffold thinking, not provide answers
- Too much assistance (from parents or AI) can weaken independence
- AI is dangerous when it replaces effort, but valuable when it supports it
- The best help is often invisible to the learner
- Frustration is normal and can be productive if managed correctly
- Children learn more when they believe they solved the problem themselves
- AI should act like a background assistant, similar to his own parenting approach
- Learning outcomes are emotional as well as academic

DOES

Observations: What did this person DO that surprised you?

- Completed his child's LEGO project overnight, then intentionally undid it so the child could rebuild it herself
- Structured the work to make success possible without giving answers directly
- Avoids AI on homeworks if they can
- Encouraged the child to step away when frustrated instead of pushing through immediately
- Chose to support learning indirectly rather than intervening live
- Did not introduce AI tools, despite being aware of them
- Avoided setting formal AI rules, relying instead on personal values about learning
- Positioned himself as a background assistant, not the primary problem-solver
- Prioritized the child's confidence and independence over speed or completion
- Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Worried when his child becomes frustrated or wants to quit
- Skeptical and uneasy about AI replacing real learning
- Proud when his child succeeds independently
- Relieved when taking a break helps reset frustration
- Protective of his child's confidence and self-esteem
- Torn between wanting to help and not wanting to interfere
- Cautiously optimistic about AI when it is used appropriately
- Frustrated by tools or systems that encourage shortcuts
- Satisfaction from helping in a way that preserves independence
- Responsible for guiding learning without being overly controlling

Person B

SAYS

Observations: What did this person SAY that surprised you?

- "It has to be double checked"
- "(AI's) sometimes faster than google"
- "I don't like AI note takers. Annotating it yourself helps retention."
- "It can't generate anything new."
- "More practice problems would be helpful.
- "Usually uses AI for complicated definitions… to simplify explanations.""
- "I don't trust AI, it hampers critical thinking skills."
- "Flashcards are okay. It's simple enough not to mess up."
- "AI will take over jobs filled with busy work."
- "I'm against overuse… especially for homework assignments like essays."

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks AI is helpful, but should not replace thinking
- Thinks others using AI raises the standard for everyone else
- Automation weakened retention
- Students often rely on AI due to heavy workloads
- People use AI just to how quickly it can get work done
- Students assume AI is correct too quickly. Not double checking it
- Learning can only be done through engagement + critical thinking
- Entry level workers are the most likely to be replaced by ai
- AI can only form an result but not understanding
- AI should only be used a support tool

DOES

Observations: What did this person DO that surprised you?

- Uses AI while studying/practicing
- Played with their hair, pondering future usage
- Avoids AI on homeworks if they can
- Uses AI as a faster search engine
- Actively double checks all the results from AI
- Still prefers textbook examples and explanations
- Avoids AI even on things such as lectures and notes
- Will still go to officer hours before AI explanations if possible
- Played with their phone and hair when discussing how they used AI
- Searches for additional practice in various

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Anxious about the future job loss
- Anxious about being better than AI in order to get hired
- Conflicted about its efficiency and its impact on learning
- Tired of competing w/ others who use AI
- Reflective, thinking about new tools including AI
- Skeptical of the accuracy of AI
- Angered by the efficiency and performance of students using GPT
- Uneasy about the normalization of AI
- Conflicted about the classes that encourage the usage of AI
- Upset at the difficulty of avoiding AI's usage

Person C

SAYS

- "I usually start by reading documentation and examples."
- "I learn best by prototyping and implementing things myself."
- "AI is useful as a starting point, not as a primary source."
- "I don't blindly trust AI outputs."
- "Explaining my thinking to coworkers helps uncover logical flaws."
- "AI works well for general problems, not specific domains."
- "If a tool isn't useful, I just don't use it."
- "Learning at work requires both high-level understanding and detailed application."
- "AI should be a force multiplier, not something you rely on."
- "AI shouldn't be added to problems that don't actually need it."

THINKS

- Deep understanding comes from doing, not just reading or generating answers.
- Trust must be earned through citations and verifiable sources.
- Over-reliance on AI can stunt professional growth.
- AI hallucinations are a real risk in specialized technical work.
- Learning is part of the job, not separate from it.
- Validation of information is a critical learning step.
- Tools should support thinking, not replace it.
- Productivity gains don't automatically mean better learning.
- Moderation is key when using AI.
- Not all AI integrations are beneficial or ethical.

DOES

- Reads official documentation when learning new technology.
- Builds prototypes to explore how tools actually work.
- Searches forums and existing solutions online.
- Asks coworkers for feedback or second opinions.
- Explains problems aloud to others to refine thinking.
- Uses AI early to get orientation or keywords
- Stops using AI once problems become domain-specific
- Cross-checks AI outputs against trusted sources.
- Iterates from prototype to production-ready solutions.
- Selectively adopts tools based on usefulness, not hype.

FEELS

- Cautiously optimistic about AI's potential.
- Skeptical of AI outputs without sources.
- Confident in hands-on learning approaches.
- Concerned about misinformation and hallucinations.
- Motivated to grow professionally without shortcuts.
- Neutral rather than excited about AI adoption.
- Frustrated by poorly integrated or unnecessary AI features
- Empowered when AI improves efficiency *without* replacing thinking.
- Responsible for maintaining learning quality.
- Wary of AI tools that encourage passive acceptance.

Person D

SAYS

Observations: What did this person SAY that surprised you?

- "I give them hints but not detailed solutions"
- "For abstract mathematics questions, I discuss a concrete example"
- "AI is beneficial as long as it's used wisely"
- "They may use AI for references, but they must engage in critical thinking"
- "They should be able to complete closed-book exams independently"
- "Teachers should be aware of AI tools and help students use them appropriately"
- "Some schools are reluctant to adopt AI"
- "There are many more open-access publications now"
- "I support AI use, but it shouldn't replace thinking"
- "There are many more open-access journals and resources now than before""

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks real understanding comes from working through difficulty
- Giving answers too quickly hurts long-term learning
- Thinks AI can support learning, but it can't replace reasoning
- Thinks students must maintain intellectual independence
- Over-reliance on AI weakens mastery
- Educators should shape how AI is used not ignore it
- Struggle is a necessary part of graduate education
- Efficiency is less important than depth of understanding
- Students should be able to perform without external tools
- Academic rigor must be protected as technology evolves

DOES

Observations: What did this person DO that surprised you?

- Intentionally avoid giving full solutions
- Guides by pointing key observations
- Relies on examples to help build intuition
- Shares own graduate school experiences when advising
- Allows struggle before stepping in
- Allows AI usage only within clear boundaries
- Emphasizes independent problem solving
- Encourages to verify information themselves
- Trusts traditional exams to test understanding
- Mentors informally rather than micromanaging

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Cautiously optimistic about AI's role in education
- Protective of academic standards
- Responsible for modeling good learning practices
- Concerned about students becoming dependent on tools
- Confident in traditional teaching methods
- Measured and deliberate when discussing AI
- Slightly frustrated by institutional hesitation around AI
- Calm about technological change, but vigilant
- Motivated to preserve deep learning
- Trusting of students, but aware of potential shortcuts

Person E

SAYS

Observations: What did this person SAY that surprised you?

- "She's pretty independent at this stage and manages most of her academics on her own"
- "Her studying is very self-directed"
- "It's less homework and more constant studying and revision"
- "The material is extremely content-heavy"
- "She usually tries to work through things herself first"
- "Sometimes she just vents when the workload feels overwhelming."
- "It's demanding and stressful, but also very structured."
- "Time management and burnout are the biggest challenges."
- "At first, I was unsure about AI because accuracy really matters in medicine."
- "I see AI as more of a supplement - like a quick tutor"

THINKS

Inferences: Based on what they SAY and DO, what might they be THINKING?

- Thinks deep learning still requires effort and repetition
- Thinks AI can help with understanding but not mastery
- Accuracy matters more than speed in medicine
- Burnout is a constant risk
- Studying well is about structure and discipline
- Short explanations can save time and reduce frustration
- Real learning can't be outsourced
- Efficiency is helpful but only if learning quality stays high
- Support systems matter even for independent learners
- AI should feel like tutoring not automation

DOES

Observations: What did this person DO that surprised you?

- "She studies for long hours reviewing lectures and materials"
- "She relies heavily on practice questions and question banks."
- She uses textbooks and lecture slides before outside tools
- She reaches out to classmates or professors when stuck.
- She explains concepts out loud to process them.
- She uses AI to clarify or simplify difficult topics.
- She avoids using AI as a shortcut.
- She manages most studying independently.
- She vents stress verbally rather than asking for answers
- She balances multiple resources instead of relying on one

FEELS

Inferences: Based on what they SAY and DO, what might they be FEELING?

- Overwhelmed by workload intensity
- Stressed but highly motivated
- Reassured by structured routines
- Frustrated by time pressure
- Cautiously optimistic about AI
- Relieved when concepts are clarified quickly
- Mentally fatigued from constant studying
- Confident in her independence
- Supported by having someone to talk to
- Anxious about burnout and long-term sustainability

HMW 1

Color label tasks (green/yellow/red)

Ask students why they are using AI before giving help

Unlock answers only after students attempt the task first

Prevent AI help on tasks the mimic exams

Built in reflection prompts. Go back to check retention/summarize

Let students witch between study, practice, or submission modes

Require Students to explain concepts before AI provides answers

Should we just disable AI as a whole to force people to go back to age w/o AI. We could totally pull a 30k and erase all existence of AI

Offer prompts designed to promote real understanding

Show AI's confidence or uncertainty for each response

Only provide citations, not full explanations

Ask students to relect on what they learned after using AI

Let professors set rules on how AI can be used

Give a summary of how AI was used for transparency

Adjust AI support based on workload or signs of burnout

HMW 2

Gate kept solutions, must explain / send in a pdf before extra info given

Utilize a bunch of active call back techniques. Learning, questions, summaries, etc.

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Require an attempt by the student first. ONLY GIVE PARTIAL HINTS otherwise

Built in reflection prompts. Go back to check retention/summarize

Have an AI that double checks your reasoning

INCLUDE REFLECTIONS, require an exit ticket type of answer

Should we just disable AI as a whole to force people to go back to age w/o AI. We could totally pull a 30k and erase all existence of AI

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Explain it back features?

For mental aspect maybe include progress? Encourage learning

Reassurance measures. Let them know they aren't behind

Allowed the child to believe the success was entirely her own, reinforcing confidence rather than revealing his involvement

Maybe let the person control strictness?

Modes of restrictions to allow learning?

HMW 3

Forces users to attempt before progressing, only give them piece by piece information.

Before the AI gives a hint, the user must explain their current understanding in their own words; the AI then adapts hints based on gaps in reasoning.

Instead of answering questions, the AI identifies and challenges hidden assumptions in the user's approach ("What are you assuming about X here?").

An AI that never explains directly, but instead guides users to the most relevant sections of official docs and asks questions that help interpret them.

Built in reflection prompts. Go back to check retention/summarize

AI only activates after a task is completed, helping users reflect on decisions, trade offs, and alternative approaches they could explore next time.

Hints are time-locked or attempt-locked; the AI nudges users to keep working before offering help, reinforcing persistence.

The AI remains silent until the user explicitly logs what they tried, what failed, and why — only then does it offer directional feedback.

The AI helps by adding constraints rather than removing them. For example: "Solve this without using X," or "What if X increased by tenfold?

Instead of giving correct solutions, the AI generates plausible wrong approaches and asks the user to critique them.

Limited hints given per problem, e.g. 3 tries before the user is on their own

Instead of giving answers, search the most relevant tutorials/learning resources available.

Once assistance is given for a certain type of problem, no assistance is allowed on another of the same type.

The AI asks users to rate their confidence before receiving help and adjusts its responses accordingly. lower confidence = broader guidance, higher confidence = sharper challenges.

Lightweight AI that only gives limited hints. Further requests are denied and links to other resources are given